Supplementary Material: VoxClinBench: A Cross-Lingual, Cross-Disease Clinical Voice Biomarker Benchmark

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This supplement expands the main paper along five axes: training configuration (S1), per-seed raw numbers with bootstrap confidence intervals (S2), Griffin–Lim bridge details (S5), the subject-ID audit for the LODO depression experiments (S6), and expanded negative results (S7). Datasheet, task card, benchmark-scope comparison, submission protocol, licence table, fairness and multiple-testing details follow. All numbers trace to artifacts under `artifacts/modal_sync/` in the anonymized release.

1 Training Hyperparameters and Modal Configurations

All experiments ran on Modal (workspace name anonymized, max 10 concurrent GPUs). Scripts and per-experiment YAMLs are under `scripts/`. A `volume.commit()` checkpoint is saved after every stage so any run is resumable from partial results.

1.1 8-Modality CNN-Transformer

Setting	Value
Branches (modalities)	spec, mfcc, mel, ppg, ema, prosodic, static, wavlm_l9
Per-branch encoder	4-layer 1D CNN (64/128/256/256 ch) $\rightarrow$ 2-layer Transformer (d=256, h=4)
Fusion	Concatenate pooled tokens, 2-layer MLP head, 12 binary sigmoids
Parameters	36,785,928
Optimizer	AdamW, lr $= 1\times10^{-4}$, weight decay 1×10^{-4}
LR schedule	Cosine with 5-epoch linear warmup, 60 epochs total
Batch size	16 (subject-level), 2 gradient-accum steps
Loss	BCEWithLogits, per-disease `pos_weight` $= N_{\text{neg}}/N_{\text{pos}}$ (train set)
Seeds	$\{0, 1, 2, 3, 4\}$ for Tier-2 fair-eval; best-val epoch selected per seed
Early stop	Keep epoch with highest macro-AUROC on val set
GPU	T4 ($0.59/hr), $\sim$2.1 h per seed

1.2 Frozen SSL Probes (NeuroVoz, SVD, Depression LODO)

Setting	Value
Backbones	WavLM-Base, HuBERT-Base, wav2vec2-Base (L2/L5/L9/L12); XLSR-53 (L6/L12/L18/L24)
Feature extraction	30 s mean-pooled hidden state, batch=24, no gradients, A10G
Probe	`sklearn.LogisticRegression`, L2 penalty, $C = 1.0$, `class_weight="balanced"`
Pre-processing	`StandardScaler` fit on train fold only, applied to val/test
Split	Subject-level stratified 80/20, 3 seeds $\{42, 1, 2\}$ (SVD, NeuroVoz) or $\{0, 1, 2\}$ (depression)
Bootstrap CI	2000 resamples of held-out subjects, percentile 2.5/97.5
GPU/Cost	Extraction T4 or A10G; LogReg on CPU (negligible)

1.3 LP-FT Schedule

Two-phase schedule used for the SVD and NeuroVoz LP-FT bake-offs (Tables 2 and 1). The B2AI v0.2 wav2vec2 / HuBERT full-FT runs used a shorter **3 LP + 3 FT** schedule owing to per-task small-N early-stop pragmatics (6–48 positives per task); full-FT collapse values (mean 0.45–0.56 across backbones $\times$ diseases) are archived under `artifacts/modal_sync/v0_2_upgrade/eval_b2ai_{wav2vec2,hubert}_ft/`.

Phase	Configuration
Phase 1 (LP)	Frozen backbone, linear head, lr $= 1 \times 10^{-3}$, AdamW, 20 epochs
Phase 2 (FT)	Unfreeze full backbone, lr $= 1 \times 10^{-5}$ (backbone) / 1×10^{-4} (head), 10 epochs
AMP	`torch.amp.autocast("cuda")` + `GradScaler`; `scaler.scale(loss).backward()` $\rightarrow$ `unscale_` $\rightarrow$ `clip_grad_norm_(1.0)` $\rightarrow$ `scaler.step(opt)` $\rightarrow$ `scaler.update()`
Batch	2, grad-accum 4, clip 15 s waveforms
GPU	A10G (22 GB, \$1.10/hr), $\sim$14–16 GB VRAM utilized

Hyperparameter provenance. The LP-FT recipe above (LR pair $(10^{-5}, 10^{-4})$, AdamW, warmup, AMP, two-phase schedule) follows SUPERB [17] and LP-FT [11] verbatim; batch/clip are VRAM-driven. We fixed the recipe *a priori* rather than grid-searching on the test split. The same recipe yields 0.958 LP-FT AUROC on NeuroVoz and 0.893 on SVD (Tables 1, 2); B2AI full-FT collapse is a small-N failure mode ($n_{\text{pos}} \in [38, 73]$), not under-tuning. Alternative recipes may be submitted via `voxbench compare`.

1.4 Modal Deployment Table

Experiment family	Script	GPU	Timeout
WavLM extraction (cached features)	`eval_b2ai_multilayer_ssl.py`	A10G	3 h
SVD SSL bake-off	`eval_svd_ssl.py`	A10G	4 h
NeuroVoz SSL bake-off	`eval_neurovoz_ssl.py`	A10G	4 h
B2AI AVQI/clinical feat.	`eval_b2ai_avqi.py`	T4	2 h
B2AI multilayer SSL	`eval_b2ai_multilayer_ssl.py`	A10G	6 h
8-mod CNN-Trans.	`train.py` + `marvel/`	T4	6 h
AVEC multilayer F1	`eval_avec_multilayer.py`	A10G	4 h

All entries use Modal volume `b2ai-voice-data`.

1.5 Cost Log (approximate, rounded to nearest $)

Experiment family	GPU-hours	Approx. $
8-modality CNN-Transformer (2 seeds $\times$ 12 disease runs $\times$ 9 LOO)	38	22
SVD SSL bake-off (4 backbones $\times$ 3 seeds + LP-FT)	9	10
NeuroVoz SSL bake-off (4 backbones $\times$ 3 seeds + LP-FT)	8	9
Depression LODO (3 holdouts $\times$ 3 seeds $\times$ 4 backbones)	12	13
B2AI multilayer SSL (16 backbone $\times$ layer $\times$ 3 diseases $\times$ 5 seeds)	18	20
AVQI/clinical (T4)	3	2
AVEC multilayer F1	6	7
Total direct compute	**94**	**$\sim$83**

Figures exclude one-off extraction passes and failed preflight runs (<$0.50 each). All detached jobs (`modal run -d`) were preflighted for $\geq$1 min without `-d` to verify data-loading, first-batch loss decrease, and dependency resolution before submission.

1.6 Per-Baseline Reproducibility

The table below documents, for every reference baseline cited in the main paper, the seed set, Modal GPU class, wall time per seed, estimated cost, and the key optimizer / schedule hyperparameters. All 5-seed baselines use the fixed seed list $\{0, 1, 2, 3, 4\}$; 3-seed baselines use $\{42, 1, 2\}$ (inherited from the MARVEL reference [15] to preserve comparability). All training runs that update transformer weights use mixed-precision (`torch.amp.autocast("cuda")`) paired with `GradScaler`; without `GradScaler`, FP16 gradients underflow to zero and loss stays flat, so the pairing is non-optional in our harness. Every Modal training job writes a `code_snapshot/` directory into both the Modal volume (`b2ai-voice-data`) and the local `artifacts/modal_sync/<run>/` mirror, so that any numerical re-check can diff the exact code revision against the current tree before supplementing seeds (see our data-integrity protocol in supplementary §2). Wall-time cells are the actual per-seed container wall time reported by the training scripts; cost is $\$/\text{hr}_{\text{GPU}} \times$ wall at the Modal 2026-Q1 rate card (T4: $0.59; A10G: $1.10; A100: $3.06). Cells marked "—" reflect that the run aggregated multiple seeds or multiple layers inside a single container and the per-seed wall time was not separately logged; the total container wall time is cited in footnote in those rows.

Baseline	Seeds	GPU	Wall / seed	$ / seed (tot.)	Key hparams (bs / lr / ep. / pat. / optim.)	Artifact
MARVEL (WavLM-Large probe + LogReg, seasoned rerun)	{42,1,2}	A10G[a]+T4	55 min[b]	$1.01 ($3.0)	LogReg head; stage-1 encoder 14 ep, stage-2 14 ep; AdamW $1e{-}4$, wd $1e{-}5$, cosine; bs 108; pat. 10	`seasoned_leakfree_20260407/t2_seed_{42,1,2}_`
8-modality CNN–Transformer (v0.2 headline)	{0,1,2,3,4}	A10G	$\approx$72 min[b]	$1.32 ($6.6)	bs 108; AdamW $1e{-}4$, wd $1e{-}5$; cosine $T_{\max}{=}40$, $\eta_{\min}{=}1e{-}6$; max 40 ep; pat. 10; focal+LS; pos-weighted BCE	`v0_2_upgrade/eval_b2ai_5seed_rerun/results_5seed.json`
WavLM-Base L9 frozen probe (VoxClinBench-Base)	{0,1,2,3,4}	A10G	—[c]	$1.20 ($6.0)	Feature extraction (no grads), $batch{=}64$ utterances; subject-level mean-pool; LogReg 80/20, `class_weight=balanced`, `C=1.0`; StandardScaler per-seed	`b2ai_multilayer_ssl_20260412/`
wav2vec2-base full FT (LP→FT)	{0,1,2,3,4}	A10G	$\approx$45 min[b,c]	$0.83 ($4.1)	bs 2, grad-accum 4 (eff. 8); 15 s audio; LP 3 ep (head only, AdamW $1e{-}3$); FT 3 ep (backbone $1e{-}5$, head $1e{-}4$); wd $1e{-}4$; AMP+GradScaler	`v0_2_upgrade/eval_b2ai_wav2vec2_ft/results_wav2vec2_ft.json`
HuBERT-base full FT (LP→FT)	{0,1,2,3,4}	A10G	$\approx$45 min[b,c]	$0.83 ($4.1)	bs 2, grad-accum 4; 15 s; LP 3 ep / FT 3 ep; LR schedule as above; AMP+GradScaler	`v0_2_b2ai_hubert_ft_20260413/results_hubert_ft.json`
Whisper-large-v3 frozen encoder probe	{0,1,2,3,4}	A10G	—[c]	$1.80 ($9.0)	Encoder-only, no grads; extract layers {8,16,24,32}; $batch{=}16$ utterances; per-seed 80/20 LogReg; `class_weight=balanced`	`v0_2_upgrade/eval_b2ai_whisper_encoder/results_whisper.json`
XLSR-53 L18 probe (SVD / NeuroVoz external)	{42,1,2}	A10G	—[c]	— ($$\sim$9)[e]	Frozen feature extraction bs 12; LogReg + LP-FT head; SEEDS hard-coded in `eval_svd_ssl.py`; backbone dim 1024	`svd_ssl_20260412/part1_xlsr53_l18.json`, `part2_seed{42,1,2}.json`

[a] MARVEL stage-1 extraction runs on A10G (`marvel_faithful_multiseed.py` L273–274, `gpu="A10G"`, 2 h timeout); the stage-2 seasoned head-training step runs on T4 (L282–283, `gpu="T4"`, 2 h). [b] Per-seed wall time estimated from Modal `timeout` ceiling divided by `n_seeds` for the run that completed at that ceiling, cross-checked against the `SWEEP-BUDGET` comment in `scripts/eval_single_mod_t4_v2.py` L4 ("baseline=15 min/seed expected=$4.5 kill_at=$10") after normalizing for 12-disease vs. single-disease runs. Actual `wall_min` was not persisted in the result JSONs for these runs; see INTEGRATION doc for the forward-fix. [c] Frozen-encoder probes amortize feature extraction across seeds in a single container; per-seed split is not meaningful (LogReg fit is <1 s per seed on CPU). Cell shows total container cost instead. [d] LP-FT scripts set `gpu="A10G"` and `timeout=60*60*3` in `scripts/eval_b2ai_hubert_ft.py` L219–220 and the matching wav2vec2 script; per-seed estimate is 5-task $\times$ (LP 3 ep + FT 3 ep) on A10G at the measured 2 batches/min for FT per our Modal runtime table. [e] XLSR-53 row shares a single extraction pass with the other SVD-SSL backbones (wavlm, wav2vec2, hubert); the $9 figure in supplementary §2 is the joint cost of the 4-backbone $\times$ 3-seed bake-off and is not split per backbone.

Checkpoint locations. All Modal checkpoints live under the volume path `/data/checkpoints/<run>/` with a parallel local mirror at `artifacts/modal_sync/<run>/` (pulled via `modal volume get b2ai-voice-data /checkpoints/<run>`). Each run-directory contains: (i) `results_*.json` (mean/std and per-seed AUROCs), (ii) `predictions/` (per-sample probabilities used for bootstrap CIs), (iii) `code_snapshot/` (the exact `train.py` / `marvel/` sources used for that run), and, for fine-tuning runs, (iv) `state_dict.pt` checkpoints written every 5 epochs and on best-validation improvement.

Preflight discipline. Every detached Modal submission (`modal run -d`) is preceded by a non-detached preflight of $\geq$1 min that verifies data loading, first-batch loss decrease, and dependency resolution. Preflight artifacts are written to `/data/checkpoints/<run>/_preflight/` (not the main checkpoint dir) to prevent the resume-logic taint described in our Modal protocol.

2 Full Per-Seed AUROC Tables

Every headline number in the main paper is broken out here with per-seed AUROC and 95% bootstrap confidence interval (2000 resamples of held-out subjects).

2.1 B2AI 8-Modality, 11 Tier-2 Diseases, 5 Seeds (v0.2)

Source: `artifacts/modal_sync/v0_2_upgrade/eval_b2ai_5seed_rerun/results_5seed.json`. 5-seed subject-level rerun $\{0, 1, 2, 3, 4\}$, test-set AUROC at best-val epoch per seed. Psychiatric-history excluded per Tier-2 protocol (§3).

Disease	Mean ± SD (5 seeds)	95% bootstrap CI	Notes
VF paralysis	0.959 ± 0.016	[0.905, 0.998]	
Laryngeal dystonia	0.947 ± 0.013	[0.854, 0.998]	
Cognitive impairment	0.913 ± 0.013	[0.855, 0.962]	
Airway stenosis	0.916 ± 0.006	[0.830, 0.980]	
Parkinson's	0.874 ± 0.015	[0.800, 0.935]	
Benign lesions	0.834 ± 0.030	[0.691, 0.950]	
Depression	0.700 ± 0.125	[0.511, 0.885]	high seed variance
PTSD	0.707 ± 0.033	[0.580, 0.822]	
MTD	0.703 ± 0.028	[0.553, 0.840]	small-N (n_+=58); see §7.1
ADHD	0.619 ± 0.060	[0.484, 0.747]	
Chronic cough	0.525 ± 0.022	[0.336, 0.700]	small-N (n_+=41); see §7.1
Psychiatric history	0.708 ± 0.037	[0.591, 0.815]	
macro-physical-8	**0.834 ± 0.008**	[0.791, 0.873]	primary physical-family headline
macro-psychiatric-4	**0.684 ± 0.061**	[0.553, 0.805]	primary psychiatric-family headline
macro-12 (combined)	**0.784 ± 0.020**	[0.731, 0.831]	secondary; reported for continuity
macro-11 (legacy, psychiatric_history excluded)	0.791 ± 0.019	—	superseded by dual-family macros

CIs are per-subject stratified bootstrap, B=2000 per seed, averaged across 5 seeds. Source: per-subject prediction CSVs pulled from Modal volume `b2ai-voice-data:checkpoints/tier2__v0_2_seed{0..4}/analysis/patient_predictions_tier2_unified.csv`; computation in `scripts/compute_bootstrap_ci_v0_2.py`; summary at `artifacts/modal_sync/v0_2_bootstrap_preds/bootstrap_ci_summary.json`. Minor mean drift ($\leq$ 0.01) vs. the `results_5seed.json` table arises because eval-time AUROC uses recording-level aggregation while here we compute at the patient-level consistent with per-subject bootstrapping.

The pre-v0.2 2-seed MARVEL-vintage table has been removed; its 0.858 macro is superseded by the 5-seed patient-level values above. See §6 for retraction context.

2.2 NeuroVoz SSL Bake-Off, 3 Seeds

Source: `artifacts/modal_sync/neurovoz_ssl_20260411/results.json`. NeuroVoz [12, 13] (Spanish Parkinson's, CC BY-NC-ND 4.0 via Zenodo 10.5281/zenodo.10777657): 108 subjects (53 PD / 55 HC), 2,903 recordings, 80/20 subject-level, seeds $\{42, 1, 2\}$.

Backbone (frozen LogReg)	Seed 42	Seed 1	Seed 2	Mean $\pm$ SD
WavLM-Base L9	1.000	0.964	0.900	0.955 $\pm$ 0.041
HuBERT-Base L9	1.000	0.982	0.900	0.961 $\pm$ 0.043
wav2vec2-Base L9	0.955	0.982	0.882	0.939 $\pm$ 0.042
XLSR-53 L18	0.982	0.991	0.891	0.955 $\pm$ 0.045

Adaptation (WavLM)	Seed 42	Seed 1	Seed 2	Mean $\pm$ SD
Frozen	1.000	0.964	0.900	0.955 $\pm$ 0.041
Full FT	0.955	0.927	0.809	0.897 $\pm$ 0.063
LP-FT	1.000	0.964	0.909	0.958 $\pm$ 0.037

Observation: backbone differences (0.939–0.961) are within one seed-SD of each other; HuBERT's 0.006 lead over WavLM/XLSR-53 is inside noise. The decisive effect is the adaptation mode: full FT degrades by -0.058 mean.

2.3 SVD SSL Bake-Off, 3 Seeds

Source: `artifacts/modal_sync/svd_ssl_20260412/results.json`. SVD Rekurrensparese [10] (German VF paralysis; Saarbrücken Voice Database, Institute of Phonetics, Universität des Saarlandes): 213 patients + 300 randomly-subsampled healthy controls, 80/20 subject-level, seeds $\{42, 1, 2\}$.

Backbone (frozen LogReg)	Seed 42	Seed 1	Seed 2	Mean $\pm$ SD
WavLM-Base L9	0.906	0.894	0.855	0.885 $\pm$ 0.022
HuBERT-Base L9	0.933	0.914	0.883	0.910 $\pm$ 0.024
wav2vec2-Base L9	0.928	0.901	0.856	0.895 $\pm$ 0.034
XLSR-53 L18	**0.947**	**0.933**	**0.907**	**0.929 $\pm$ 0.021**

Adaptation (WavLM)	Seed 42	Seed 1	Seed 2	Mean $\pm$ SD
Frozen	0.906	0.894	0.855	0.885 $\pm$ 0.022
Full FT	0.758	0.740	0.710	0.736 $\pm$ 0.022
LP-FT	0.915	0.902	0.862	0.893 $\pm$ 0.022

XLSR-53's +0.019 to +0.044 margin over English-only SSL is outside one SD: language-matched SSL pre-training (XLSR-53 includes $\sim$2000 h of German MLS) dominates scale.

2.4 HiFi-GAN Vocoder Spike (v0.2, 5 seeds)

Source: `artifacts/modal_sync/v0_2_upgrade/eval_b2ai_hifigan_spike/results_hifigan_spike.json`. WavLM-base L9 frozen mean-pooled probe on Griffin–Lim-reconstructed vs. HiFi-GAN-vocoded B2AI, three weak phonation conditions.

Disease	GL (baseline)	HiFi-GAN
Benign lesions	0.501 $\pm$ 0.017	0.494 $\pm$ 0.076
MTD	0.560 $\pm$ 0.031	0.603 $\pm$ 0.084
Chronic cough	0.594 $\pm$ 0.063	0.519 $\pm$ 0.078

HiFi-GAN does not reliably improve over GL: only MTD gains ($+0.04$), benign lesions is flat, chronic cough *degrades* (-0.08). A neural vocoder upgrades perceptual quality but does not restore the cycle-to-cycle phase signal either. See main §6(c).

Table 1: NeuroVoz Parkinson's disease (Spanish; $N = 108$ subjects, 2,903 recordings). Subject-level AUROC under within-dataset subject-stratified $80/20$ splits, 3 seeds, $\pm$ std shown. Frozen = logistic-regression probe on mean-pooled SSL features; LP-FT = linear-probe-then-fine-tune; Full FT = end-to-end fine-tuning. †MARVEL within-NeuroVoz trained on spec+MFCC is a supplementary reference point derived from cached predictions; see KEY §3.1 entry for NeuroVoz within.

Model	**AUROC**
HuBERT L9 (frozen)	**0.961±0.043**
WavLM L9 (frozen)	0.955±0.041
XLSR-53 L18 (frozen)	0.955±0.045
wav2vec2 L9 (frozen)	0.939±0.042
MARVEL (spec+MFCC)†	0.935±0.042
MARVEL zero-shot (B2AI-¿NeuroVoz)	0.528
LP-FT WavLM	0.958±0.037
Full FT WavLM	0.897±0.063

3 Cross-Lingual Transfer: NeuroVoz Parkinson's (Spanish)

We ask whether the Parkinson's acoustic signature transfers from English (B2AI) to Spanish (NeuroVoz). Within-dataset 80/20 splits over 3 seeds (Table 1; per-seed numbers in §2.2) show that frozen SSL probes dominate: HuBERT L9 [9] reaches 0.961 ± 0.043, the strongest single model, with WavLM L9 [4] at 0.955, XLSR-53 L18 [5] at 0.955, and wav2vec2 L9 [1] at 0.939—all four within seed noise. Linear-probe-then-fine-tune (LP-FT) matches the frozen baseline at 0.958, whereas full fine-tuning degrades performance to 0.897 (-0.058), echoing the invariance-rewriting pattern we observe on depression. Cross-domain transfer from B2AI to NeuroVoz using logistic-regression probes is harsher: only EMA (0.628) and WavLM (0.667) exceed chance; a WavLM+EMA ensemble reaches 0.683, and a full ensemble combining the WavLM probe with PPG and EMA classifiers trained on B2AI climbs to 0.850. Few-shot adaptation on NeuroVoz with frozen WavLM L9 reaches 0.881 at K=5 and 0.942 at K=20 per class, while MARVEL [15] zero-shot transfer (B2AI→NeuroVoz) reaches 0.528. The PD motor signature is evidently dominated by features that survive language change, and same-architecture SSL probes exceed the MARVEL baseline's zero-shot AUROC by over 0.43; this is a directional observation on a single corpus.

4 Cross-Lingual Transfer: SVD Vocal Fold Paralysis (German)

Our third cross-lingual study targets unilateral vocal-fold paralysis on the German SVD corpus (213 Rekurrensparese vs. 300 healthy speakers, 80/20 splits, 3 seeds; per-seed in §2.3). To our knowledge this is the first systematic SSL backbone bake-off for German voice pathology. Under a frozen-probe logistic-regression protocol (Table 2), XLSR-53 L18 [5] posts the highest single-baseline AUROC at 0.929 ± 0.021, followed by HuBERT L9 [9] at 0.910, wav2vec2 L9 [1] at 0.895, and WavLM L9 [4] at 0.885. The gap XLSR-53 opens over WavLM here ($+0.044$) is striking because on NeuroVoz (Spanish) the same four backbones all fall within seed noise at 0.94–0.96. The mechanism we propose is straightforward: XLSR-53 was pretrained on roughly 2000 hours of German audio from MLS, while HuBERT, wav2vec2 and WavLM are English-only (LibriSpeech 960h). Language-matched SSL pretraining is therefore the dominant factor for cross-lingual voice-pathology detection once the target language drifts far from English. LP-FT holds at 0.893, but full fine-tuning drops to 0.736 (-0.15 AUROC), the largest full-FT drop we observe in any cross-lingual cell. We also attempted joint B2AI+SVD training to exploit shared motor pathology; the domain gap between Griffin-Lim-reconstructed B2AI audio and raw SVD audio made mixed training *worse* than separate training on each corpus (§7 subsection on joint training); we report this as an empirical observation on this configuration.

Table 2: SVD vocal fold paralysis (German; 213 Rekurrensparese vs. 300 healthy, $N = 513$ subjects total). Subject-level AUROC under within-dataset subject-stratified $80/20$ splits, 3 seeds, $\pm$ std shown. Language-matched XLSR-53 pretraining shows $+0.02$–0.04 higher AUROC than English-only SSL backbones on this corpus.

Model	**AUROC**
XLSR-53 L18 (frozen, multilingual incl. $\sim$2k h DE)	**0.929±0.021**
HuBERT L9 (frozen, EN)	0.910±0.024
wav2vec2 L9 (frozen, EN)	0.895±0.034
WavLM L9 (frozen, EN)	0.885±0.022
LP-FT WavLM	0.893±0.022
Full FT WavLM	0.736±0.022

5 Griffin–Lim Bridge Details

5.1 Configuration

The B2AI v3.0 public release provides spectrograms, not raw audio. We invert spectrograms to pseudo-waveforms via Griffin–Lim iteration before feeding any SSL model.

Parameter	Value
Sample rate	16 000 Hz
FFT size (`n_fft`)	400
Hop length	160 (10 ms)
Window length	400 (25 ms)
Window type	Hann
Iterations (CPU, `librosa.griffinlim`)	32
Iterations (GPU, `torchaudio.transforms.GriffinLim`)	60
Power	2.0 (magnitude^2)
Source $\rightarrow$ output	B2AI derived spectrogram $\rightarrow$ 16 kHz mono waveform

Cosine-similarity calibration. On DAIC-WOZ/E-DAIC (both raw and derived spectrograms available), mean cosine between WavLM-L9 embeddings on raw vs. GL-reconstructed waveforms is 0.979 (n=189) / 0.975 (n=275). High-dimensional SSL embeddings concentrate on a low-rank cone (arbitrary-clip cosine ≥ 0.9), so these prove only that *SSL-relevant* invariances survive GL, not phase-faithfulness. Phase-dependent features (jitter, shimmer, CPPS, AVQI) collapse on GL audio (§7.2): GL is a valid input for SSL probes, not for phase-dependent extractors.

6 LODO Subject-ID Audit and DAIC-WOZ Retraction

Headline finding (2026-04-14). A subject-ID intersection on the participant-ID metadata TSVs shipped with each corpus shows that DAIC-WOZ (IDs 300–492, n=189) is a **strict subset** of E-DAIC (IDs 300–718, n=275): all 189 of 189 DAIC-WOZ IDs appear in E-DAIC. DAIC-WOZ is therefore *not* listed as a separate benchmark corpus in this release (v0.3); the v0.2 DAIC-WOZ task (A15) and v0.2 LODO row with DAIC-WOZ as test corpus (B3) are **retracted**. The E-DAIC row (A16a, n=275) is kept and now subsumes the 189-subject DAIC-WOZ cohort plus the 86 additional AVEC 2019 interviews. In a companion decision (2026-04-14, *partially reverted* 2026-04-14), **MODMA is retained as a cross-lingual validation target only** (N=52 Chinese, zero-shot, directional claim; sits outside the Holm family). The v0.2 MODMA row (A17, 0.799 ± 0.016) is kept in main Table 1 with the training pool relabelled "E-DAIC n=275 EN" (DAIC-WOZ $\subset$ E-DAIC adds no unique subjects, so the prior "DAIC+E-DAIC" pool is duplicate-free under this relabel; the value is numerically unchanged). The v0.2 MODMA-*in-training* LODO rows (B2, B4, B5) remain retracted because they relied on the DAIC+E-DAIC duplication. Numerically retracted: (i) 8-mod DAIC-WOZ LOO AUROC 0.910; (ii) MARVEL-adv DAIC-WOZ LOO 0.825 ± 0.028; (iii) all MODMA-*in-training* LODO cross-training numbers (B2/B4/B5) and any row that used DAIC-WOZ as a separate training or test corpus. The E-DAIC within-dataset value reported in main Table 1 (A16a, 0.600 ± 0.075) has

no MARVEL comparator at this release because all published MARVEL E-DAIC values came from DAIC-contaminated LODO.

Corpus	Release / Composition
DAIC-WOZ	AVEC 2016/2017 release, n=189 participants, USC virtual agent "Ellie"
E-DAIC	AVEC 2019 release, n=275 participants, *extended* with new interviews
MODMA (target-only)	Lanzhou University dataset, n=52 Chinese participants; retained as cross-lingual *validation target only* (never a training source in v0.3 LODO); n=52 limits claims to directional validation, not ranking
B2AI v3.0	Bridge2AI Voice Consortium, US multi-site, disjoint from AVEC

B2AI disjointness. B2AI (US English) shares no participants with AVEC (DAIC-WOZ / E-DAIC) by construction (different countries, different IRBs, different recruitment); the ID schemes are mutually exclusive namespaces. Disjointness from MODMA is guaranteed by construction (Chinese participants recruited at Lanzhou University versus US English-speaking B2AI participants).

7 Failed Experiments (Expanded): Negative Results on Weak B2AI Diseases

This section expands the negative-results discussion of the main paper (summarised in the main paper's Additional Results subsection and Discussion). Two independent strategies—hand-crafted clinical perturbation features, and a 16-way multilayer SSL bake-off—converge on a 0.49–0.73 AUROC plateau for benign lesions, MTD, and chronic cough. The v0.3 paired GL-vs-raw causal study (§7.1) shows this is small-N (30–48 positives per task on test fold), *not* a Griffin–Lim phase-loss artifact. We also document a failed joint-training attempt (B2AI+SVD) that motivates our per-corpus ensemble design.

7.1 GL-vs-raw Causal Study (v0.3)

Sources: `artifacts/modal_sync/v0_3/eval_edaic_gl_vs_raw/results.json` and `artifacts/modal_sync/v0_3/eval_svd_gl_vs_raw/results.json`. WavLM-Base L9 frozen mean-pooled probe trained on identical subjects, audio range, and seeds, swapping only raw-vs-GL audio. 5 seeds $\{0, 1, 2, 3, 4\}$, subject-disjoint 80/20.

Dataset (task)	N	n_+	raw mean±SD	GL mean±SD	Δ	paired p
E-DAIC (depression)	275	66	0.600 ± 0.075	0.618 ± 0.071	$+0.018$	0.080
SVD (voice pathology)	513	213	0.831 ± 0.026	$\mathbf{0.863 \pm 0.010}$	$\mathbf{+0.032}$	$\mathbf{< 0.001}$

E-DAIC per-seed AUROC. raw: $\{0.731, 0.542, 0.625, 0.516, 0.584\}$; GL: $\{0.749, 0.590, 0.608, 0.537, 0.604\}$; Δ per seed: $\{+0.018, +0.048, -0.016, +0.020, +0.020\}$, mean $+0.018 \pm 0.020$.

SVD per-seed AUROC (Rekurrensparese vs Gesund, healthy sub-cap 300). raw: $\{0.800, 0.843, 0.802, 0.867, 0.841\}$; GL: $\{0.866, 0.865, 0.870, 0.870, 0.844\}$; Δ per seed: $\{+0.067, +0.021, +0.068, +0.003, +0.003\}$, mean $+0.032 \pm 0.029$, paired bootstrap $p < 0.001$.

Conclusion. Across two domains — prosodic (depression) and phase-sensitive (vocal-fold paralysis / pathological phonation) — GL matches or modestly exceeds raw on the WavLM-L9 probe. The earlier "Griffin–Lim phase-loss ceiling" framing is therefore retracted; the B2AI plateau on benign lesions, MTD, and chronic cough is attributable to small-N ($n_+ \in [41, 58]$) rather than reconstruction artifact. Hand-crafted phase-sensitive features (jitter, shimmer, AVQI; §7.2) still collapse on GL audio because they directly measure cycle-to-cycle phase, but SSL-pooled embeddings are robust.

7.2 AVQI and Clinical Acoustic Features on GL Audio

Source: `artifacts/modal_sync/b2ai_avqi_20260412/results.json`. n=762 subjects, 5 seeds $\{0, 1, 2, 3, 4\}$, subject-level 80/20, `StandardScaler` + LogReg, 2000-resample bootstrap.

Disease	Feature subset	n_{feat}	AUROC (mean ± SD)	95% CI
Benign lesions (n_+=53)	all	62	0.488 ± 0.069	[0.344, 0.651]
	AVQI-only	2	0.553 ± 0.044	[0.348, 0.735]
	jitter+shimmer	16	0.497 ± 0.074	[0.318, 0.603]
	cough	34	0.452 ± 0.069	—
MTD ($n_+ \approx 58$)	all	62	0.513 ± 0.066	[0.32, 0.72]
	AVQI-only	2	0.599 ± 0.025	[0.41, 0.85]
Chronic cough ($n_+ \approx 41$)	all	62	0.579 ± 0.041	[0.31, 0.84]
	jitter+shimmer	16	0.630 ± 0.047	[0.49, 0.78]

Every subset sits at or below the 8-mod CNN-Transformer baseline (0.694/0.733/0.633). Because jitter, shimmer, CPPS, and AVQI are phase-dependent fine-grained perturbation measures, the Griffin–Lim phase reconstruction dominates the measurement and masks the disease signal *for these features specifically*. This is a downstream-task null for hand-crafted phase-perturbation extractors on GL audio; SSL probes do *not* share this null — see §7.1 where a matched WavLM-L9 probe matches or exceeds raw audio on both prosodic (E-DAIC) and phase-sensitive (SVD) tasks.

7.3 Joint B2AI + SVD Training

Source: `artifacts/modal_sync/joint_training_20260411/results.json`. VF Paralysis, WavLM-L9 features, LogReg. B2AI uses GL→WavLM; SVD uses raw→WavLM.

Training corpus	B2AI test AUROC	SVD test AUROC
B2AI only	0.698	0.461
SVD only	0.445	0.900
Combined	0.628	0.878

Mixed training is *worse* than either corpus alone on its own held-out set: GL-reconstructed features $\neq$ raw-audio features, and the pooled optimizer compromises between two modes it should not average. For cross-dataset SVD/B2AI transfer, joint training underperforms per-corpus training on both held-out sets in this configuration; we report the negative result without prescribing an alternative.

8 Fairness Analysis (Preliminary)

We report a preliminary subgroup fairness analysis on B2AI v3.0.0. The probe is a single-modality WavLM layer-9 frozen-feature LogReg classifier, evaluated with 5-seed 80/20 splits over 762 subjects using features cached from the multilayer SSL run. Subgroups are defined along sex (M, F, unknown), age (<30, 30–50, 50–70, >70), and language. All 762 subjects are labeled `unknown` for language in the current v3.0.0 demographics export, so no cross-lingual subgroup comparison is possible here; we defer this to the planned v3.1 release.

Honest caveat. The single-modality L9 probe is markedly weaker than our 8-modality main baseline, with overall AUROC in the 0.45–0.55 range for many conditions. At this operating point, per-subgroup AUROC deltas are dominated by seed noise and small-n variance rather than genuine demographic disparity.

We flagged subgroup deltas with $|\Delta\text{AUROC}| > 0.10$ versus the overall mean: vocal fold paralysis, sex=M, Δ=+0.133 (n_{pos}=18); benign lesions, age>70, Δ=$-$0.144 (n_{pos}=5); ADHD, age 50–70, Δ=+0.101 (n_{pos}=20); depression, age 50–70, Δ=$-$0.121 (n_{pos}=5). All four flagged deltas have $n_{\text{pos}} \leq 20$ and fall within the 2000-bootstrap CI of the overall estimate.

A full 8-modality fairness analysis is deferred to future work. The current preliminary analysis is insufficient to detect or exclude demographic disparity, and we flag this explicitly as a limitation. Raw per-subgroup numbers: `artifacts/modal_sync/b2ai_fairness_20260412/results.json`.

9 Multiple Testing Correction

The main paper reports twelve exceedance/tie/first-of-kind claims (one per Tier 2 condition). To guard against the multiple-comparisons problem, we apply Holm–Bonferroni correction to the subset of comparisons that admit a valid within-cohort baseline (MARVEL on B2AI, or a published same-dataset SOTA). Under the family decomposition specified in the main paper (§3, Correction), Family A (physical) contributes $m{=}8$ within-cohort B2AI tasks against MARVEL and Family B (psychiatric) contributes $m{=}0$ (no within-cohort MARVEL comparator at this release; all Family B entries are descriptive). Per-test α under Bonferroni with Family A size $m{=}8$ and family-wise $\alpha{=}0.05$ is $0.05/8 \approx 0.0063$; Family B is reported descriptively with no Holm correction, and these are what the main paper *Exceeds* verdict is evaluated under. In v0.2 the paired p-values are computed via the DeLong U-statistic test [6, 14] (`voxbench compare --test delong`) using the fast $O(N \log N)$ Sun–Xu kernel; the subject-level paired bootstrap ($B{=}2000$) remains available via `--test paired-bootstrap` as a nonparametric cross-check. The DeLong implementation ships with 7 unit tests verifying agreement with reference SciPy/pROC values and recovering known-direction cases.

Condition	Same-cohort MARVEL?	8-mod 5-seed patient-level mean ±SD	MARVEL 3-seed mean ±SD	Holm–Bonferroni p
Parkinson's	yes	0.874±0.015	0.873±0.019	n.s.
Airway stenosis	yes	0.916±0.006	0.909±0.014	n.s.
VF paralysis	yes	0.959±0.016	0.844±0.033	see §2
Laryngeal dystonia	yes	0.947±0.013	0.820±0.056	see §2
Cognitive impairment	yes	0.913±0.013	0.927±0.025	n.s.
Benign lesions	yes	0.834±0.030	0.771±0.050	see §2
MTD	no	0.703±0.028	—	descriptive
Chronic cough	no	0.525±0.022	—	descriptive
Depression	no	0.700±0.125	—	descriptive
PTSD	no	0.707±0.033	—	descriptive
ADHD	no	0.619±0.060	—	descriptive
Psychiatric history	no	0.708±0.037	—	descriptive

Retractions and environment addenda. An earlier 2-seed draft with inflated point estimates and an untraceable MARVEL comparator column is retracted; the 5-seed table above supersedes. AUPRC for chronic cough / ADHD (low-prevalence) will be added in the camera-ready per-seed release. Artifact paths: feature store `/data/b2ai_voice_v3.h5` on Modal volume `b2ai-voice-data`; per-modality parquets at `/data/external/analysis/*/features_*.parquet` (use `sparc_pitch.parquet`; `torchaudio_pitch.parquet` is corrupted). Every main-paper number traces to a JSON/CSV under `artifacts/modal_sync/`. Environment: Python 3.11, PyTorch 2.6.0 / CUDA 12.6, `transformers`≥4.40, `librosa` 0.11, `scikit-learn` 1.8, `resampy` 0.4; evaluator dependencies are lighter (NumPy/SciPy/scikit-learn/pandas) per `pyproject.toml`. Per-seed seeding via `random_state`; DataLoader shuffle seeded per experiment. Script-to-result mapping: `train.py + marvel/` for 8-modality; `scripts/eval_{b2ai_multilayer_ssl,svd_ssl,neurovoz_ssl,b2ai_avqi,modma_reverse_lodo,b2ai_fairness}`. for per-corpus re-eval.

10 Full Task Card

11 Benchmark Scope Comparison (prior benchmarks)

12 Submission Protocol (full specification)

12.1 Submission protocol

A VoxClinBench submission consists of (i) per-task, per-seed *prediction CSVs* and (ii) a *metadata manifest*. Raw audio, trained weights, and pre-processing pipelines are *not* uploaded to the leaderboard; the protocol scores model outputs, not model artefacts.

Table 3: **VoxClinBench task card (v0.2).** One row per task; 17 scored within-corpus tasks (A-family) plus 2 active cross-lingual LODO tasks (B-family). Counts $n_{\text{pos}}/n_{\text{neg}}$ are subject-level (not recording-level); recording-level totals are in supp §6. Metric column lists the primary leaderboard metric; AUPRC is additionally reported for low-prevalence ($< 10\%$) B2AI tasks. Every task uses a stratified 80/20 subject-disjoint split with a further 80/20 train/val split and 5-seed bootstrap CIs unless noted. Rows marked ‡ are retracted or dropped in v0.2; shown for ledger continuity (see supp §6). Row A17 is a zero-shot cross-lingual validation *target*, not trained on.

ID	Task (registry id)	Corpus / Lang.	Label definition	Metric	Split	$n_{\text{pos}}/$
Family A — physical / motor / laryngeal / respiratory (within-corpus)						
A1	`b2ai.parkinsons`	B2AI / en	Clinician dx (ICD-10 G20)	AUROC, AUPRC	80/20 subj., 5-seed	73/6
A2	`b2ai.airway_stenosis`	B2AI / en	iSGS clinician dx	AUROC, AUPRC	80/20 subj., 5-seed	55/7
A3	`b2ai.vf_paralysis`	B2AI / en	Vocal fold paralysis dx	AUROC, AUPRC	80/20 subj., 5-seed	60/7
A4	`b2ai.laryngeal_dystonia`	B2AI / en	Adductor spasmodic dysphonia dx	AUROC, AUPRC	80/20 subj., 5-seed	50/7
A5	`b2ai.cognitive_impair`	B2AI / en	MCI clinician dx	AUROC, AUPRC	80/20 subj., 5-seed	45/7
A6	`b2ai.benign_lesions`	B2AI / en	Benign vocal-fold lesion dx	AUROC, AUPRC	80/20 subj., 5-seed	58/7
A7	`b2ai.mtd`	B2AI / en	Muscle tension dysphonia dx	AUROC, AUPRC	80/20 subj., 5-seed	52/7
A8	`b2ai.chronic_cough`	B2AI / en	Chronic cough dx	AUROC, AUPRC	80/20 subj., 5-seed	41/7
A13	`neurovoz.parkinsons`	NeuroVoz / es	PD vs HC (clinical)	AUROC	80/20 subj., 5-seed	53/
A14	`svd.vf_paralysis`	SVD / de	Rekurrensparese (clinical dx)	AUROC	80/20 subj., 5-seed	213/
Family B — psychiatric (within-corpus)						
A9	`b2ai.depression`	B2AI / en	Depression self-report/dx	AUROC, AUPRC	80/20 subj., 5-seed	44/7
A10	`b2ai.ptsd`	B2AI / en	PTSD dx	AUROC, AUPRC	80/20 subj., 5-seed	38/7
A11	`b2ai.adhd`	B2AI / en	ADHD dx	AUROC, AUPRC	80/20 subj., 5-seed	47/7
A12	`b2ai.psychiatric_history`‡	B2AI / en	Composite self-report (dep. $\cup$ PTSD $\cup$ ADHD)	AUROC	80/20 subj., 5-seed	—
A15	`daicwoz.depression`‡	DAIC-WOZ / en	PHQ-8 ≥ 10	AUROC	pre-shipped	56/1
A16a	`edaic.depression`	E-DAIC / en	PHQ-8 ≥ 10	AUROC	pre-shipped	82/1
A16b	`edaic.phq8_regression`	E-DAIC / en	PHQ-8 continuous score	Pearson r, CCC	pre-shipped	275
A17	`modma.depression`	MODMA / zh	MDD vs HC (clinical dx)	AUROC	5-seed (single-seed src); n=52	23/
Cross-lingual LODO (leave-one-domain-out)						
B1	`lodo.pd.test_neurovoz`	B2AI-en $\rightarrow$ NeuroVoz-es	PD transfer	AUROC	zero-shot on NeuroVoz test	53/
B2	`lodo.dep.test_b2ai`‡	—	—	—	—	—
B3	`lodo.dep.test_daic`‡	—	—	—	—	—
B4	`lodo.dep.test_edaic`‡	—	—	—	—	—
B5	`lodo.dep.test_modma`‡	—	—	—	—	—
B6	`lodo.vfp.test_svd`	B2AI-en $\rightarrow$ SVD-de	Vocal fold paralysis transfer	AUROC	zero-shot on SVD test	213/

Artefact format. For each scored (task, seed) pair a submitter ships a two-column CSV

```
subject_id,predicted_prob
```

under the path convention `predictions/<task_id>.seed<s>.<submission_name>.csv`, matching the layout of the 66 reference-baseline CSVs shipped in `voxbench/predictions/` (see `INDEX.md` for the canonical file→source mapping). Ground-truth labels, demographics, and raw paths are forbidden columns and are rejected by the release filter (`voxbench/scripts/build_release.py`). A single submission is a `.zip` of this `predictions/` tree plus a `manifest.json` at the root.

Evaluation command. The harness is a pure `numpy/scipy/scikit-learn` package (no `torch` dependency) exposing a CLI identical to the reproduction surface already used for the reference baselines. The commands below describe the *reference interface* (proposed reference implementation); the shipping `voxbench` package currently exposes the entry point declaration, and the reference implementation (`voxbench/cli.py`) will land at camera-ready:

```
pip install voxbench

voxbench fetch <dataset>                    # verify credentials, print official URL + target path
voxbench eval  --task b2ai.parkinsons \
               --predictions predictions/b2ai.parkinsons.seed0.mine.csv \
               --labels your_b2ai_labels.csv
voxbench eval  --all --predictions-dir predictions/ \
               --out leaderboard_row.json   # aggregate across tasks+seeds
voxbench compare --a baseline.json --b mine.json --test delong
```

Table 4: VoxClinBench in context: scope comparison against representative prior benchmarks in general speech (SUPERB, HEAR, FLEURS, ComParE), medical imaging (MedMNIST v2), and multilingual NLP (XTREME). Columns: **Dom.** = primary domain (Speech / Med-Img / NLP); **Clin.** = clinical-disease labels; **Multi-L** = > 1 language; **X-ling.** = cross-lingual generalisation *evaluated* as a first-class task (not merely multiple monolingual tasks); **LODO** = leave-one-dataset-out / subject-disjoint cross-corpus protocol; **Harness** = code + leaderboard released. VoxClinBench is the only entry positive on **Clinical**, **Multilingual**, and **Cross-lingual transfer** simultaneously, and the only clinical-voice benchmark with an explicit LODO protocol.

Benchmark	Year	Dom.	#Tasks	#Langs	#Corpora	Clin.	Multi-L	X-ling.	LODO	Harness
VoxClinBench (ours)	2026	Speech/Clin.	**12**	**4**	**5**	✓	✓	✓	✓	✓
SUPERB	2021	Speech	10	1 (en)	∼8	✗	✗	✗	✗	✓
HEAR	2021	Audio	19	multi[†]	19	✗	✗	✗	✗	✓
FLEURS	2022	Speech	4	**102**	1	✗	✓	✓	✗	✓
ComParE	'09–'23	Paraling.	1/yr	1–2	1/yr	partial[‡]	✗	✗	✗	✓
MedMNIST v2	2022	Med-Img	18	n/a	18	✓	n/a	n/a	✗	✓
XTREME	2020	NLP	9	**40**	9	✗	✓	✓	✗	✓

[†] HEAR tasks are predominantly English-language speech plus language-agnostic music / environmental audio; HEAR does not evaluate cross-lingual transfer as a task. [‡] ComParE sub-challenges include some health-adjacent tasks (e.g. Cold, Snoring, COVID-19 cough) in individual years but are not a sustained clinical benchmark.

Labels are supplied by the submitter from the credentialed upstream corpus; the harness never sees raw audio. The evaluator recomputes AUROC, AUPRC, bootstrap CI, and Holm-corrected significance flags from the per-subject probabilities so summary-only submissions cannot game the CI width.

Metric. Primary: macro-AUROC, aggregated across the five splits seeds $\{13, 42, 2026, 7, 101\}$ with B=2000 bootstrap resamples for 95% CIs. Secondary: AUPRC for tasks with $n_{\text{pos}}/N < 0.15$; Pearson-r and CCC for the A16b PHQ-8 regression task. Between-submission comparisons use DeLong on per-subject probabilities with Holm–Bonferroni at α=0.05 within each family (A/B).

Required metadata. `manifest.json` must include: submission `name`, contact, `model_name`, `parameter_count`, SSL `backbone` (or `"none"`), training `track` (*frozen features* vs. *full fine-tune*; see §3), `license` of released weights, and a `reproducibility` field that is one of: (a) public weights URL + inference command, (b) `"weights released upon acceptance"` with an embargo date, or (c) `"closed; predictions only"`. Submissions in (c) are admitted to the leaderboard but flagged with a ‡ marker and are excluded from the *reproducible-subset* ranking.

Freeze and versioning. Splits, label definitions, and the scored task set are frozen per minor release. The current release is **v0.2**; a minor-version bump (v0.3, v0.4, . . .) is required to add or retire a scored task. Submissions made against v0.2 remain ranked under v0.2 indefinitely; the v0.3 board starts empty. A task is *deprecated* (removed from the active board but retained in the archive) if (i) the upstream corpus retracts or revokes its licence, or (ii) a train-on-test leak is documented. Deprecation carries a one-minor-version grace period before the task is pulled from the live board, mirroring the Dynabench round-lifecycle policy.

Tie-breaking. If two submissions have overlapping 95% CIs on the primary metric and a DeLong $p > 0.05$ after Holm correction, they are reported as a *tied rank band* sharing the numerically higher rank. Within a tied band, submissions are listed alphabetically by `name`; the ordering is cosmetic and carries no ranking claim.

Rejection. A submission is rejected if any of the following hold: (a) *coverage gap* — predictions are missing for more than one seed of any scored task in the declared track; (b) *metadata gap* — any required field in `manifest.json` is absent or the `reproducibility` field is not one of the three enumerated values; (c) *leakage evidence* — the submitted predictions contain subject IDs that are not in the held-out test manifest for the declared seed, or contact between a training corpus and the test set of another corpus is disclosed in the manifest. Rejected submissions are returned with a machine-readable error list so they can be resubmitted after correction.

13 Licence and Access Compliance Table

Table 5: Licence, access tier, and redistribution scope for the five corpora in VoxClinBench. All rows grounded in `artifacts/eda/{corpus}/eda_summary.json`. Under each upstream DUA we redistribute only derived artefacts (per-seed prediction CSVs, probe features where licence permits, aggregate leaderboard tables); raw recordings stay with the upstream provider.

Corpus	Language	License	Access tier	Access URL	Redistributed under DUA
B2AI	en-US	CC BY 4.0 (Bridge2AI DUA)	Registered	bridge2ai.org/voice	Features + predictions on
NeuroVoz	es-ES	CC BY 4.0	Public	zenodo.org/records/10066994	Features + predictions on
SVD	de-DE	Custom (research, registration)	Registered	stimmdatenbank.coli.uni-saarland.de	Features + predictions on
E-DAIC	en-US	AVEC DUA (restricted)†	Gated	dcapswoz.ict.usc.edu	Predictions only†
MODMA	zh-CN	Custom (Lanzhou Univ. DUA)	Registered	modma.lzu.edu.cn/data/index	Features + predictions on

† AVEC DUA restricts redistribution of raw recordings and extracted features; we ship only per-seed prediction CSVs and aggregate metrics for E-DAIC. DA

14 Datasheet for VoxClinBench

We follow the seven-section **Datasheets for Datasets** template of Gebru et al. [7]. **VoxClinBench is a benchmark harness over six existing clinical voice corpora, not a new dataset release.** We ship subject-ID manifests, evaluation code, and per-seed predictions; raw audio is fetched by the user from each upstream provider under its native license. Corpus-level provenance questions (IRB protocol, per-subject consent language, retention) are answered by each upstream corpus's own documentation; where VoxClinBench cannot independently verify an answer we write *"N/A — deferred to upstream"* rather than paraphrase. A machine-readable mirror of this datasheet is at `docs/datasheet.md` and is updated with each minor release.

14.1 Motivation

For what purpose was the dataset (benchmark) created? To provide a single subject-disjoint evaluation protocol under which published clinical voice results become directly comparable, and under which cross-lingual, cross-corpus, and cross-modality generalisation is auditable. Prior work reports results on incompatible splits, preventing apples-to-apples comparison across SSL backbones, fusion schemes, and clinical populations.

Who created the benchmark and on behalf of which entity? Anonymous for double-blind submission. Author identity, affiliation, and funding sources will be disclosed at de-anonymisation. The *upstream* corpora were created by independent consortia: Bridge2AI-Voice [2], NeuroVoz [12], SVD [16], DAIC-WOZ / E-DAIC [8], and MODMA [3].

Who funded the creation? *Benchmark harness:* disclosed at de-anonymisation. *Upstream corpora:* funding is reported in each corpus's canonical citation; we do not restate those funders here because VoxClinBench adds no new data collection.

14.2 Composition

What do the instances represent? Instances are *subject-level manifest rows* pointing to audio recordings owned by the upstream provider, together with inherited clinical or screening-instrument labels and a benchmark-assigned train/val/test fold ID. VoxClinBench never redistributes waveforms or spectrograms.

How many instances are there in total? Across the six corpora, the release covers n=2,424 unique subjects and 46,240 recordings (`artifacts/eda/dataset_summary_table.csv`): B2AI (n=833, 28,419 rec), NeuroVoz (n=108, 2,903), SVD (n=943, 13,140), DAIC-WOZ (n=189, reported upstream), E-DAIC (n=275, 275 sessions), and MODMA (n=52, 1,503). Per-task positive/negative counts appear in main Table 3.

Does the benchmark contain all possible instances, or a sample? A sample governed by three inclusion rules: (i) $n_{\text{pos}} \geq 30$ clinician-confirmed or validated labels; (ii) subject-disjoint splittability; (iii) license-compatible manifest redistribution. Tasks that fail any of these (e.g. B2AI diseases with $n_{\text{pos}} < 30$, or upstream data that forbids manifest release) are excluded. MODMA is retained as a zero-shot cross-lingual *target* only ($n=52$ keeps it outside the Holm family; see main §3).

What data does each instance consist of? *"Raw" features that VoxClinBench consumes*: B2AI v3.0 ships log-mel spectrograms (not waveforms) under PhysioNet credentialed access; NeuroVoz, SVD, DAIC-WOZ, E-DAIC, and MODMA ship raw waveforms (sample rates 8–48 kHz, median durations 3–912 s; see `eda_summary.json` per corpus). *Benchmark-added fields*: subject-disjoint split ID, task tier (T1/T2), evaluation seed, and for released runs, per-sample prediction probabilities.

Is there a label or target associated with each instance? Yes. Labels are inherited from upstream: clinician diagnosis (B2AI, NeuroVoz, SVD), PHQ-8 screening cutoff (DAIC-WOZ, E-DAIC), or HAMD/PHQ-based classification (MODMA). VoxClinBench neither re-labels nor re-adjudicates.

Is any information missing from individual instances? Demographic attributes are unevenly reported. Sex is available for NeuroVoz (0M/61F/47 missing), SVD (189M/302F/452 missing), E-DAIC (169M/105F/1 missing), and MODMA (36M/16F/0 missing); B2AI aggregate sex-by-disease is in §8. Age is reported by NeuroVoz, SVD, E-DAIC, and MODMA (means 31–68 yr; `dataset_summary_table.csv`) and by B2AI on disease subsets. Ethnicity/race is **not** uniformly reported across corpora and is *not* part of the benchmark manifest.

Are relationships between individual instances made explicit? Yes: every instance carries a subject ID, and all splits are subject-disjoint (no subject crosses train/val/test). Multiple recordings per subject are explicitly preserved in the manifest.

Are there recommended data splits? Yes, and they are *mandatory* for leaderboard submission: 80/20 subject-stratified train/test, 80/20 train/val within train, seeds $\{0,1,2,3,4\}$ for Tier-2 B2AI and supplementary full-FT baselines, $\{42, 1, 2\}$ for external-corpus bake-offs, and $\{0, 1, 2\}$ for depression LODO. B2AI E-DAIC ships with its own official train/dev/test CSVs, which we preserve.

Are there errors, sources of noise, or redundancies? Known issues documented in main §7: (i) B2AI v3.0 ships mel-spectrograms rather than waveforms, which we invert via Griffin–Lim (supp §5); the v0.3 paired GL-vs-raw causal study ($p<0.001$, 5 seeds) shows this is not the dominant error source. (ii) Depression assessment protocols differ (Ellie avatar interview vs. self-report).

Is the benchmark self-contained, or does it link to external resources? *Links to external resources.* The manifest is useless without the upstream audio, which users must fetch themselves under each provider's access regime. The `voxbench fetch` CLI verifies credentials and emits the official provider URL for each corpus.

Does the benchmark contain confidential or sensitive data? Yes: clinical diagnoses are protected health information under HIPAA / GDPR-Art. 9, and upstream corpora redistribute them only under gated access. VoxClinBench redistributes *subject IDs* + *labels* + *split assignments* under the same constraints as the upstream license. Users must hold the upstream credential to resolve IDs to audio.

14.3 Collection Process

How was the data acquired? All audio and labels were acquired by the *upstream* corpus teams under their institutional IRB / ethics approvals. VoxClinBench performs **no new data collection**. Summary of acquisition per corpus: B2AI — multi-site prospective clinical recruitment at five US sites under the Bridge2AI-Voice protocol; NeuroVoz — Hospital Universitario Gregorio Marañón, Madrid; SVD — Saarland University Hospital phoniatrics clinic; DAIC-WOZ / E-DAIC — USC Institute for Creative Technologies, Wizard-of-Oz interview protocol; MODMA — Lanzhou University MODMA protocol. Full per-subject acquisition details are **N/A — deferred to each upstream corpus's own datasheet**.

What mechanisms were used to collect the data? Hospital-grade close-talk microphones (NeuroVoz, SVD), desktop condenser microphones via the Ellie avatar interview system (DAIC-WOZ, E-DAIC), and protocol-specified hardware at B2AI sites and MODMA. *Per-device specifications are N/A — deferred to upstream.*

If the data is a sample, what was the sampling strategy? VoxClinBench applies a deterministic subject-level stratified sample: after applying inclusion rules (above), every subject in the upstream corpus that satisfies $n_{\text{pos}} \geq 30$ and is license-compatible is included. SVD healthy controls are subsampled to 300 to balance the voice-pathology task (`dataset_summary_table.csv`).

Who was involved in the data-collection process? Clinician-researchers and study coordinators at the upstream sites. Compensation details are **N/A — deferred to upstream**.

Over what timeframe was the data collected? N/A — deferred to each upstream corpus's publication: B2AI [2], NeuroVoz [12], SVD [16], DAIC [8], MODMA [3]. The VoxClinBench manifest was assembled in 2026-Q1 using the snapshot of each corpus available at that time (B2AI v3.0; NeuroVoz Zenodo 10.5281/zenodo.10777657; SVD Zenodo records 16874898 + 7024894; DAIC-WOZ and E-DAIC from `dcapswoz.ict.usc.edu`; MODMA from the Lanzhou University portal).

Were any ethical review processes conducted? *Upstream:* yes, each corpus was collected under an IRB / ethics-committee-approved protocol documented in its canonical citation. *Benchmark harness:* the present work re-uses already released data under each provider's DUA and does not involve new human-subjects collection; **specific IRB numbers are deferred to upstream corpora.**

Did you obtain consent from the individuals in question? *Upstream:* consent was obtained by the original sites under their IRB protocols. *Benchmark harness:* VoxClinBench users inherit consent scope via the upstream DUA; re-consent for downstream benchmarking is **N/A — deferred to upstream**.

Was an analysis of potential impact on data subjects conducted? Partial. We conducted preliminary fairness stratification by sex and age band on the B2AI Tier-2 tasks (supp §8), flagging disparities where detected. A corpus-wide impact assessment is an open roadmap item.

14.4 Preprocessing, Cleaning, and Labeling

Was any preprocessing / cleaning / labeling done? Yes, at two layers: *(a) Upstream preprocessing* varies per corpus and is **deferred to upstream documentation**. Notably, B2AI v3.0 ships log-mel spectrograms (40-80 bands depending on modality) rather than raw waveforms. *(b) Benchmark-level preprocessing:* Griffin–Lim inversion recovers waveforms from B2AI mel-spectrograms (supp §5); external corpora are consumed as-released. Features for SSL probes (WavLM, wav2vec2, HuBERT) are extracted from the waveform-tier input (raw or GL-reconstructed) and cached; normalisation statistics are computed *only on training folds* and applied to val/test.

Was the "raw" data saved in addition to the cleaned data? VoxClinBench does not redistribute audio, so the "raw" layer lives at the upstream provider. Per-seed extracted features, probe checkpoints, and raw predictions (`preds.npy`) are preserved in `artifacts/modal_sync/` for longitudinal audit.

Is the software used to preprocess/clean/label available? Yes. All preprocessing code ships with the release under MIT (`voxbench` PyPI package; `scripts/` directory in the code repository).

14.5 Uses

Has the benchmark been used for any tasks already? Yes: (i) the MARVEL v2 WavLM-large LogReg probe baseline, (ii) VoxClinBench-Base 8-modality CNN-Transformer, (iii) v0.3 GL-vs-raw causal study on E-DAIC and SVD, and (iv) SSL backbone bake-offs (WavLM-L9, wav2vec2, HuBERT). All numerical results are in main Table 2 with artifact paths under `artifacts/modal_sync/`.

Is there a repository linking papers / systems using the benchmark? The leaderboard and system registry will live at the release URL (disclosed at de-anonymisation).

What (other) tasks could the benchmark be used for? Speech SSL evaluation, multilingual paralinguistic transfer, fairness auditing for clinical voice AI, and vocoder / waveform reconstruction benchmarking via the GL-vs-raw protocol.

Is there anything about the composition of the benchmark or the way it was collected and preprocessed that might impact future uses? Yes, three concrete caveats: (i) **Psychiatric training is English-only in v0.3**; MODMA (n=52) is a zero-shot target, not a training source. Claims of cross-lingual psychiatric generalisation are directional, not ranking-grade. (ii) **B2AI public tier is mel-only**; waveform-dependent downstream analyses must account for the Griffin–Lim bridge. (iii) **Ethnicity/race is not available** in the manifest; users cannot audit those subgroups from VoxClinBench alone.

Are there tasks for which the benchmark should not be used? Yes, the following are explicitly out of scope, mirroring main §7: deployment screening of asymptomatic populations; employment, insurance, or immigration decisions; involuntary screening of any kind; commercial use of MODMA-derived outputs (CC BY-NC 4.0); and any clinical deployment without prospective validation and regulatory clearance.

14.6 Distribution

Will the benchmark be distributed to third parties? Yes, as a research artifact. Distribution is limited to *subject-ID manifests, evaluation code, per-seed predictions, and this datasheet*. Raw audio is **not** redistributed; the `voxbench fetch` CLI emits the official provider URL and verifies credentials.

How will it be distributed? Via (i) a PyPI package (`voxbench`) for the evaluation harness and (ii) a Zenodo-minted DOI at each tagged release. Code is mirrored on GitHub. The URL and DOI are disclosed at de-anonymisation.

When will it be distributed? v0.2 is tagged at `v0.2.0-ed`; v0.3 is scheduled for NeurIPS 2026 D&B camera-ready.

Will the benchmark be distributed under a copyright or license? Yes. Code under **MIT**; manifests, documentation, and this datasheet under **CC BY 4.0**. Each upstream corpus retains its native license (B2AI PhysioNet DUA; NeuroVoz CC BY-NC-ND 4.0 on the Zenodo record; SVD custom-research license on Saarland mirror, CC BY 4.0 on the Zenodo mirror; DAIC-WOZ and E-DAIC under USC/ICT EULA; MODMA CC BY-NC 4.0). A per-corpus license table appears in main Table 5.

Have any third parties imposed IP-based or other restrictions? Yes, via the upstream DUAs: PhysioNet credentialed access (B2AI), USC/ICT EULA (DAIC-WOZ, E-DAIC), Lanzhou University access form (MODMA, non-commercial), and Saarland registration (SVD on the native mirror). VoxClinBench cannot override these.

Do any export controls or other regulatory restrictions apply? None that we are aware of beyond standard HIPAA / GDPR protected-health-information handling, which is managed at the upstream DUA level. **Users outside the EU/US must independently verify local compliance.**

14.7 Maintenance

Who will be supporting/hosting/maintaining the benchmark? Anonymous for submission; to be disclosed at de-anonymisation together with a maintainer email and ORCID. A mirror of maintenance policy lives in `docs/datasheet.md`.

How can the owner be contacted? Maintainer email disclosed at de-anonymisation; issue tracker on GitHub at the release URL.

Is there an erratum? Yes. Breaking changes (split reshuffles, inclusion-rule changes, label corrections) bump the minor version. Prior minor-version manifests and per-seed raw predictions remain retrievable for longitudinal audit.

Will the benchmark be updated? Yes, on a roughly six-month cadence tracking upstream corpus releases (e.g. B2AI v3.0 $\rightarrow$ future v3.x). Each update is a new Zenodo-minted DOI. Leaderboard submissions are bound to the manifest version they were evaluated against.

If the benchmark relates to people, are there retention limits? Retention of the underlying audio is governed by each upstream DUA. The VoxClinBench manifest (subject IDs + labels) retains these identifiers only for as long as the upstream corpus does; if an upstream corpus retracts a subject, we issue a manifest patch within the next minor release.

Will older versions continue to be supported? Yes, by DOI. Predictions produced against an older manifest remain valid against that manifest version; leaderboard comparisons are within-version only.

If others want to extend/augment/build on the benchmark, is there a mechanism? Yes. External contributions (new corpora, new tasks, new fairness slices) are invited via GitHub pull request. Contributed tasks must satisfy the $n_{\text{pos}} \geq 30$ threshold, ship a subject-disjoint split, and arrive with a license-compatibility statement.

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